

TrES-2b

R-0072

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_191031 · created 2026-07-15T17:57:23+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458787.68575 +/- 0.00069 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.021
Transit depth (Rp/Rs)^2	1.99 +/- 0.58 [%]
Orbital Inclination (inc)	83.25 +/- 0.9
Airmass coefficient 1 (a1)	1.2921 +/- 0.0097
Airmass coefficient 2 (a2)	-0.1208 +/- 0.0026
Scatter in the residuals of the lightcurve fit is	0.74 %
Best Comparison Star	None
Optimal Aperture	5.03
Optimal Annulus	11.14
Transit Duration (day)	0.062 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.021 vs 0.1254 (deviation 0.74σ)
Duration fit vs published	0.062 d vs 0.0746 d (deviation 0.6σ)
Mid-time O–C	1.6 min
Depth SNR	6.7
Residual scatter	0.74 %

Light curve

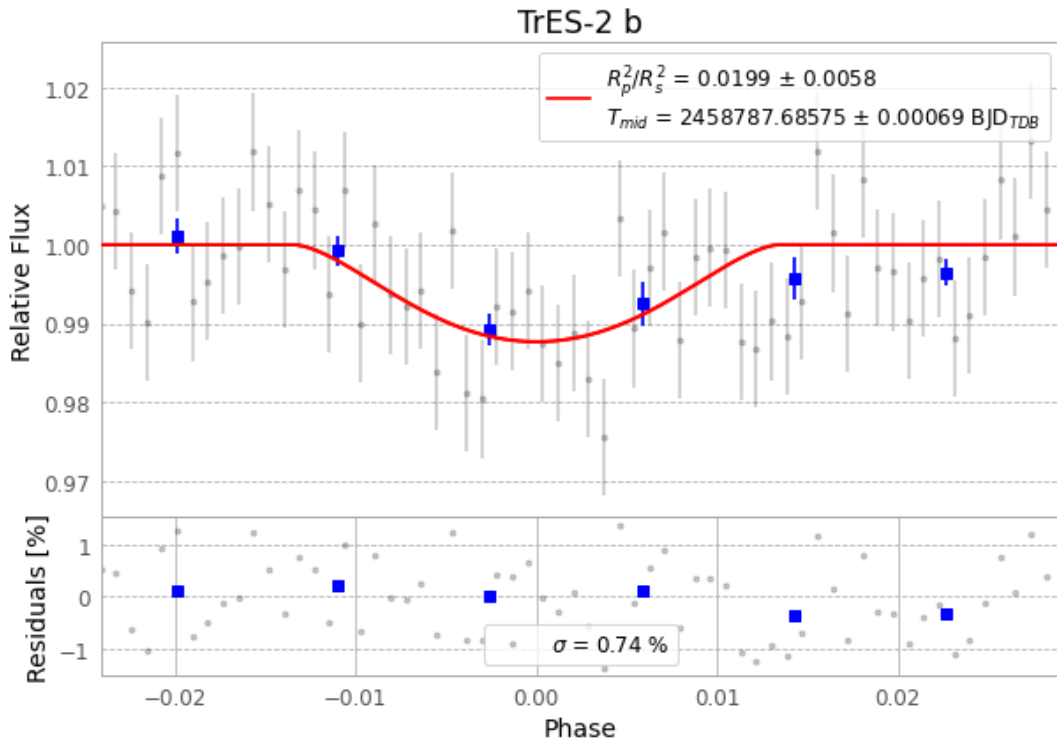


Figure 1 — FinalLightCurve_TrES-2 b_2019-10-30.png (SHA-256 364be4410b9f25a1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [268, 307], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2805576bb29f0506...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

64 FITS frames (42 MB), TRES-2191031030012.FITS → TRES-2191031060912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-191031.log (c7fbc6a6a3ce...), FinalLightCurve_TrES-2 b_2019-10-30.png (364be4410b9f...), FinalLightCurve_TrES-2 b_2019-10-30.csv (dbf678496997...)

Compute resources

Wall time 195.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 17:57Z.